

Case Study 1

By: Lindsay Gills



Part 1: Data Cleaning



Case Study: Attrition Dataset

- **Objective of Case Study:**

Identify the top three factors that contribute to turnover and build a model(s) to predict attrition and be able to measure the cost or savings impact of your model(s)

- **Executive Summary Link:**

https://docs.google.com/document/d/1fpgtkq4Q2xqpqduPtlal62mP8TA-_0T1y8d54b0HHMg/edit?usp=sharing

Data Dictionary: Attrition Dataset

Important Variables:

JobInvolvement:

Level of job involvement on a scale from 1 to 4:

- 1 = Low
- 2 = Medium
- 3 = High
- 4 = Very High

Attrition:

Indicates whether the employee has left the company ('Yes') or is still employed ('No').

MonthlyIncome:

Monthly salary of the employee.

OverTime:

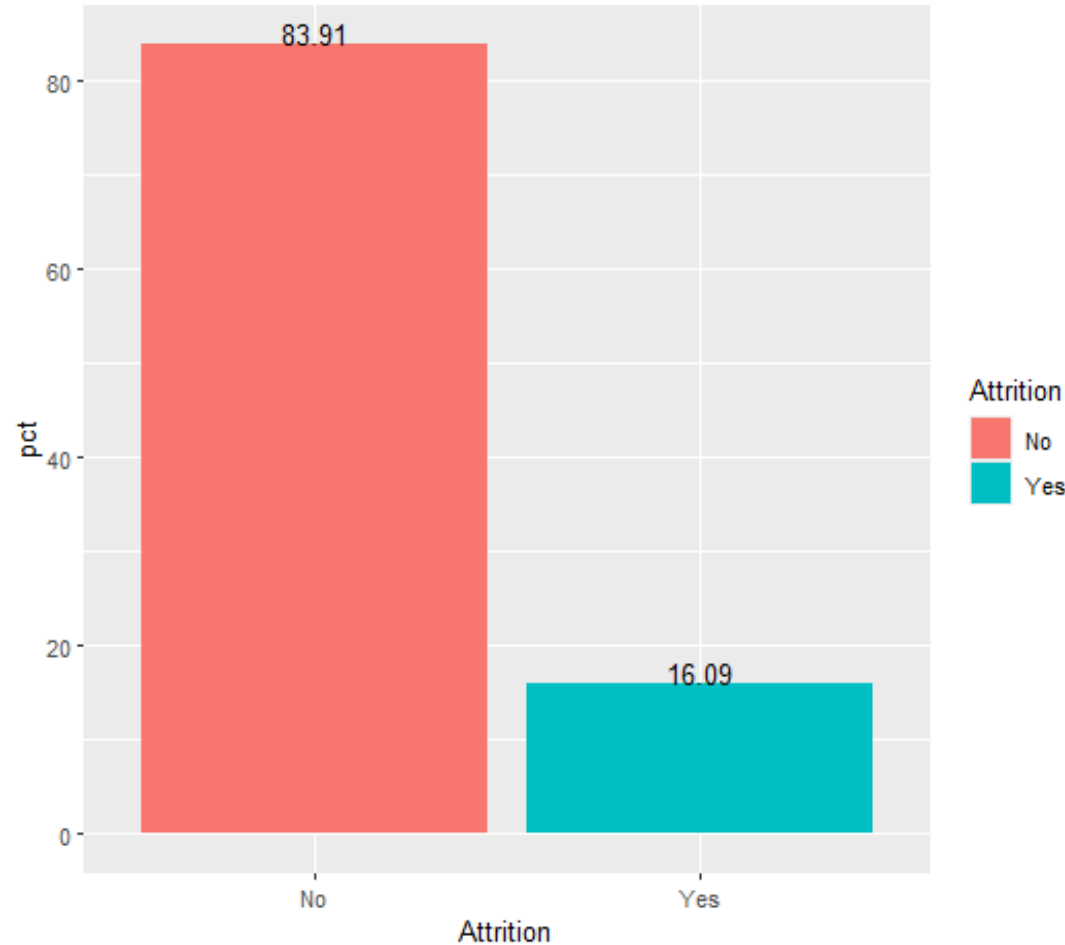
Whether the employee works overtime ('Yes' or 'No').

Null Values and Factors

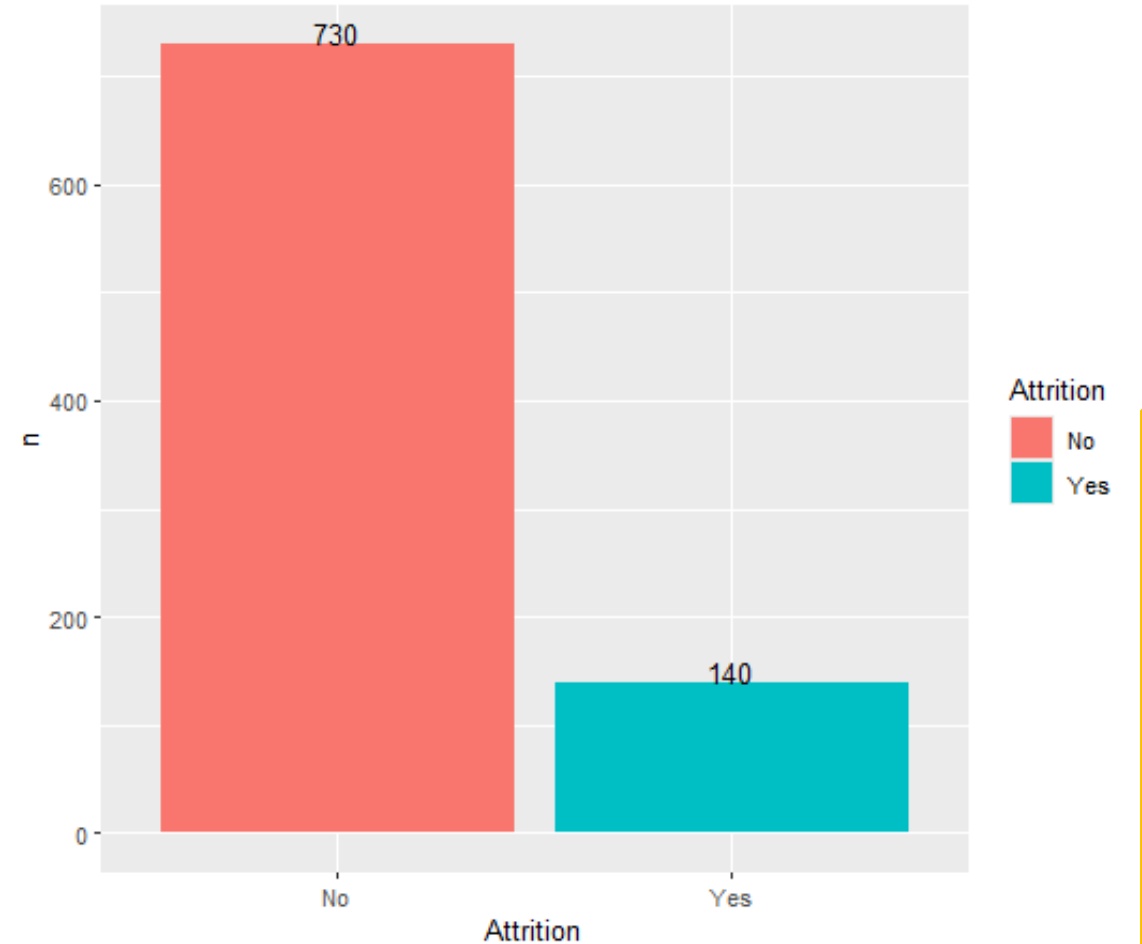
- The attrition dataset did not have any null values.
- The following columns in the attrition dataset were converted into factors:
Gender, Environment Satisfaction, Job Satisfaction, Performance Rating, Business Travel, Department, Education Field, Job Involvement, Job Level, Job Role, Marital Status, Relationship Satisfaction, Stock Option Level, Work Life Balance

Attrition and Non-Attrition Summary

Percent of Employees Who Stayed and Left The Company



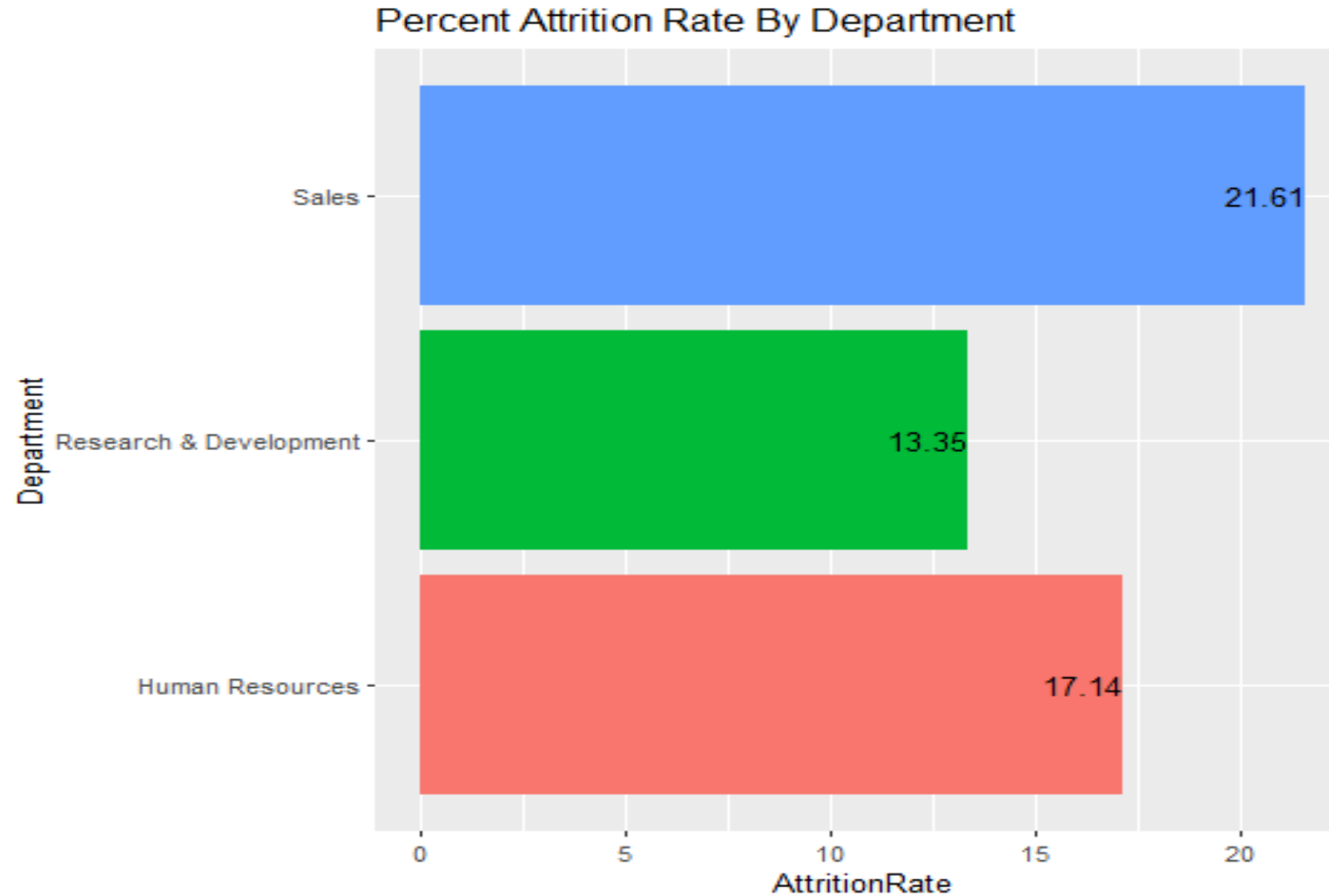
Number of Employees Who Stayed and Left The Company



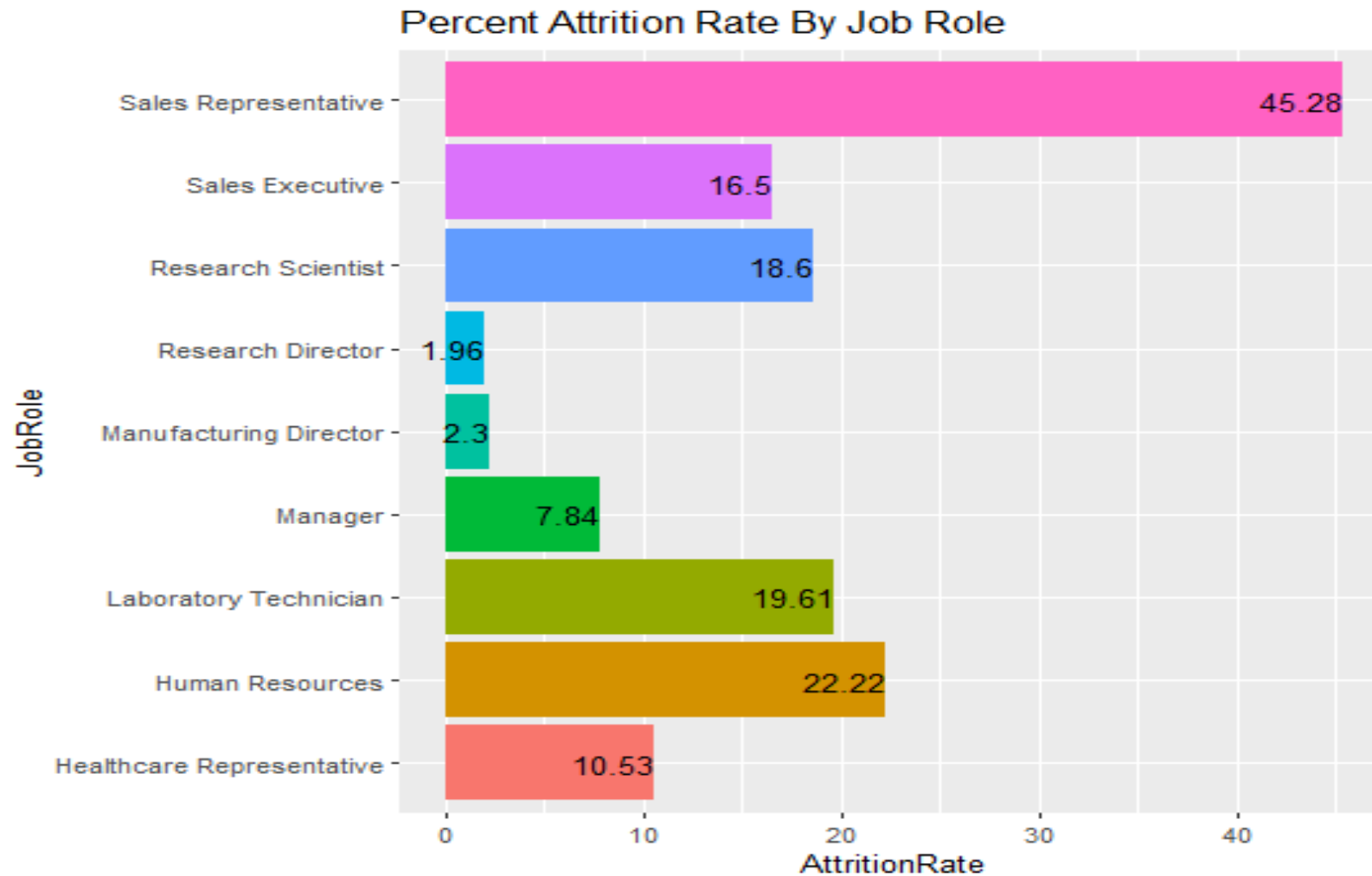
Part 2: Analyzing Data



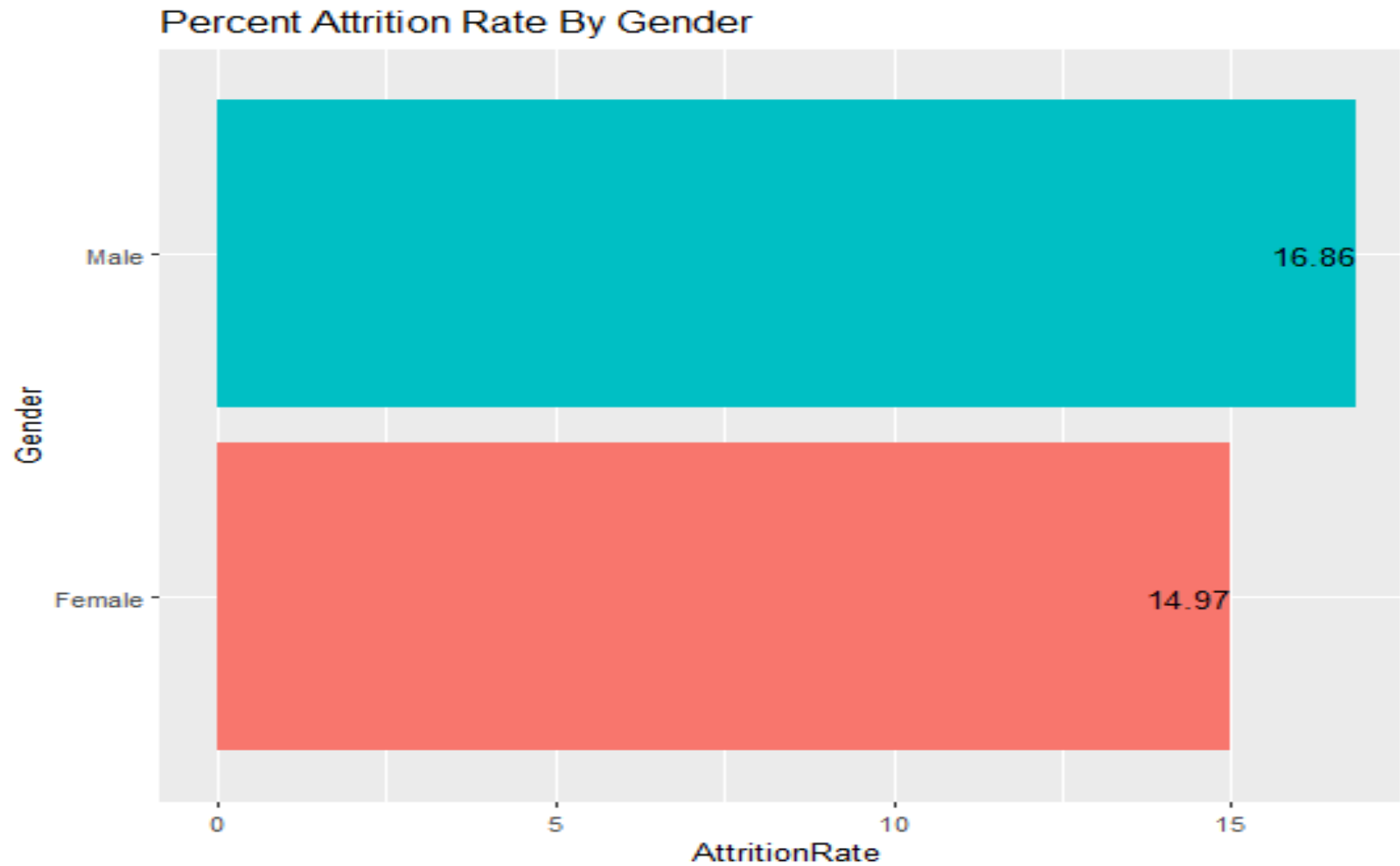
Calculating Attrition Means For Each Factor: Department



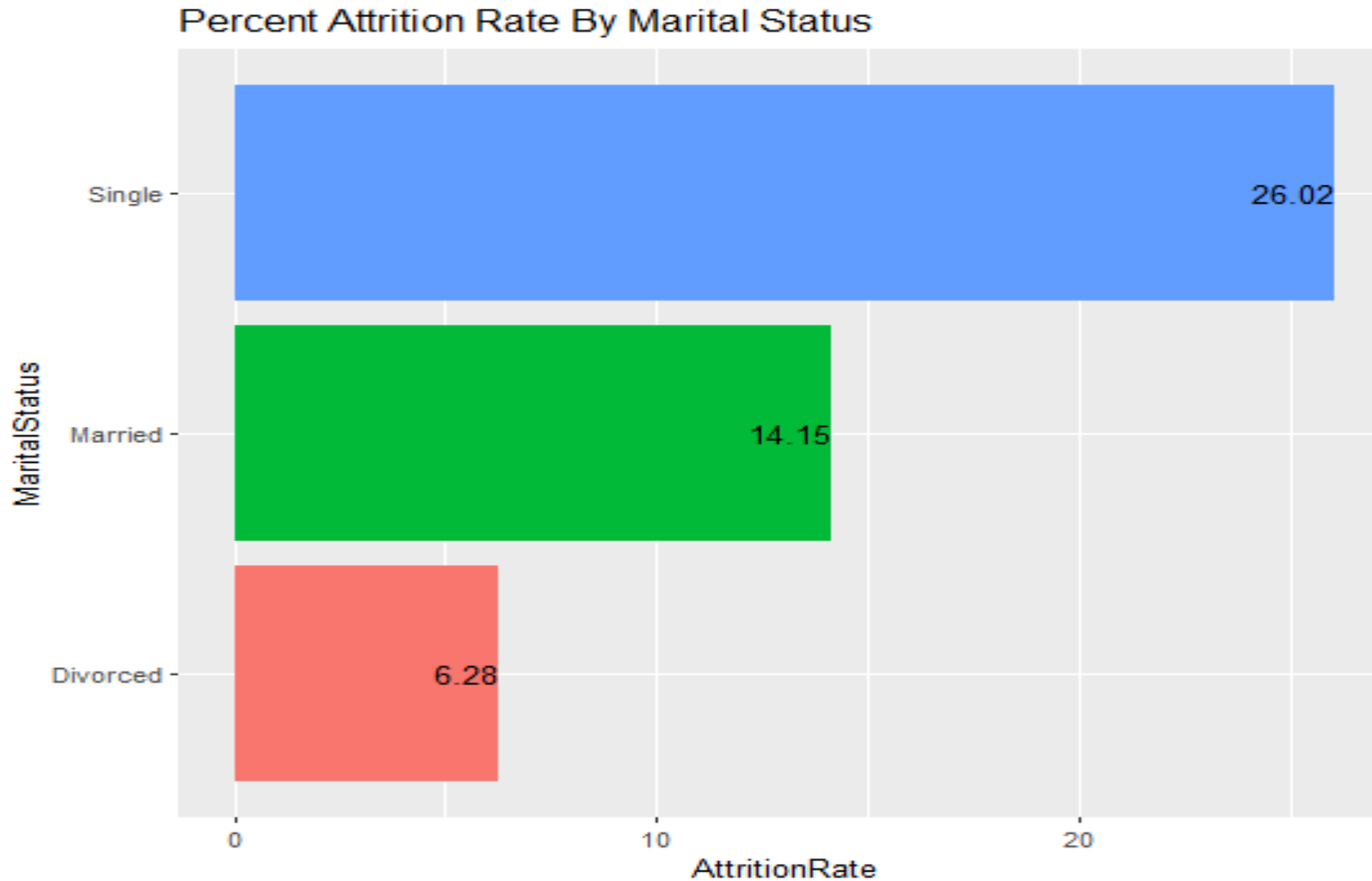
Calculating Attrition Means For Each Factor: Job Role



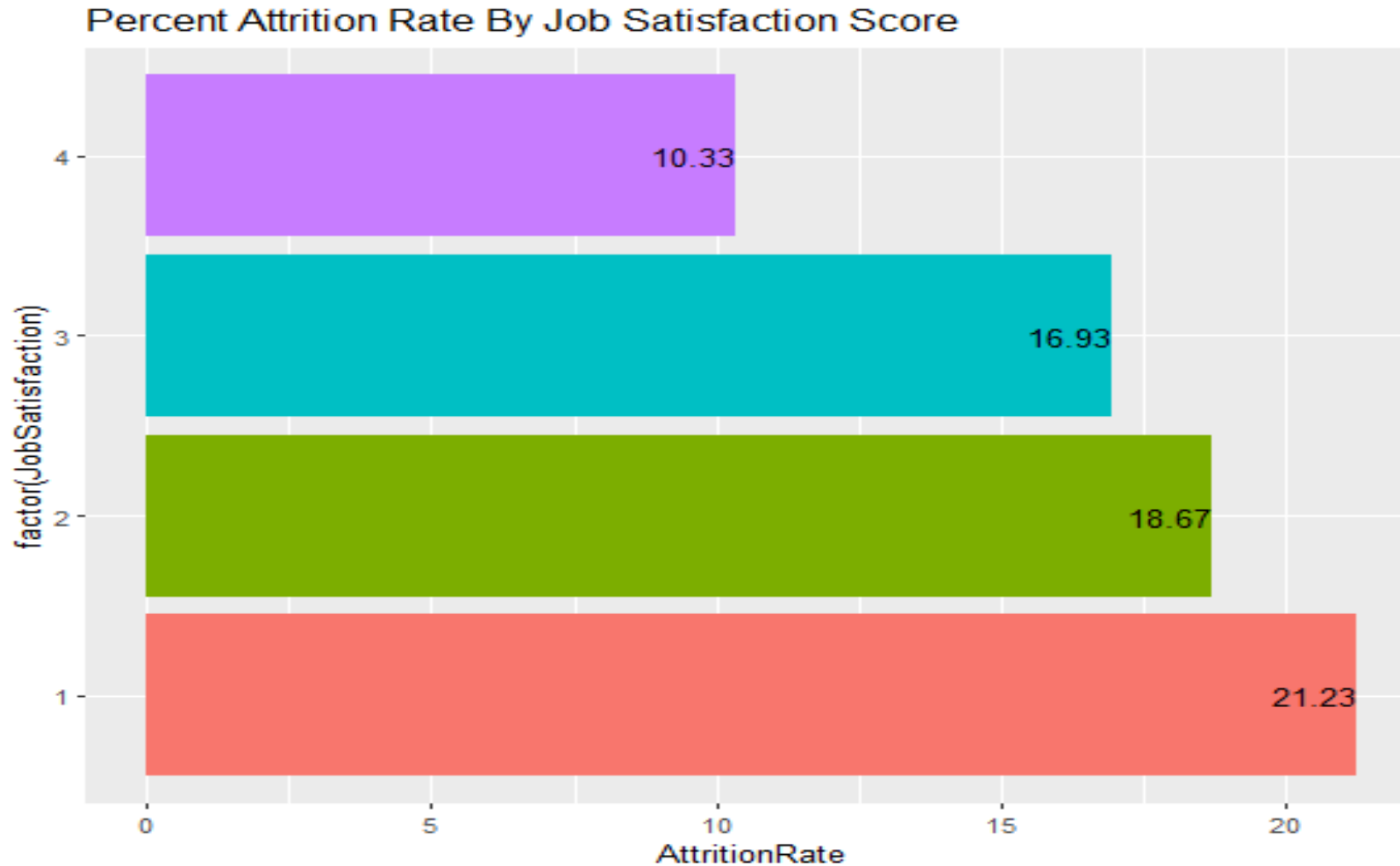
Calculating Attrition Means For Each Factor: Gender



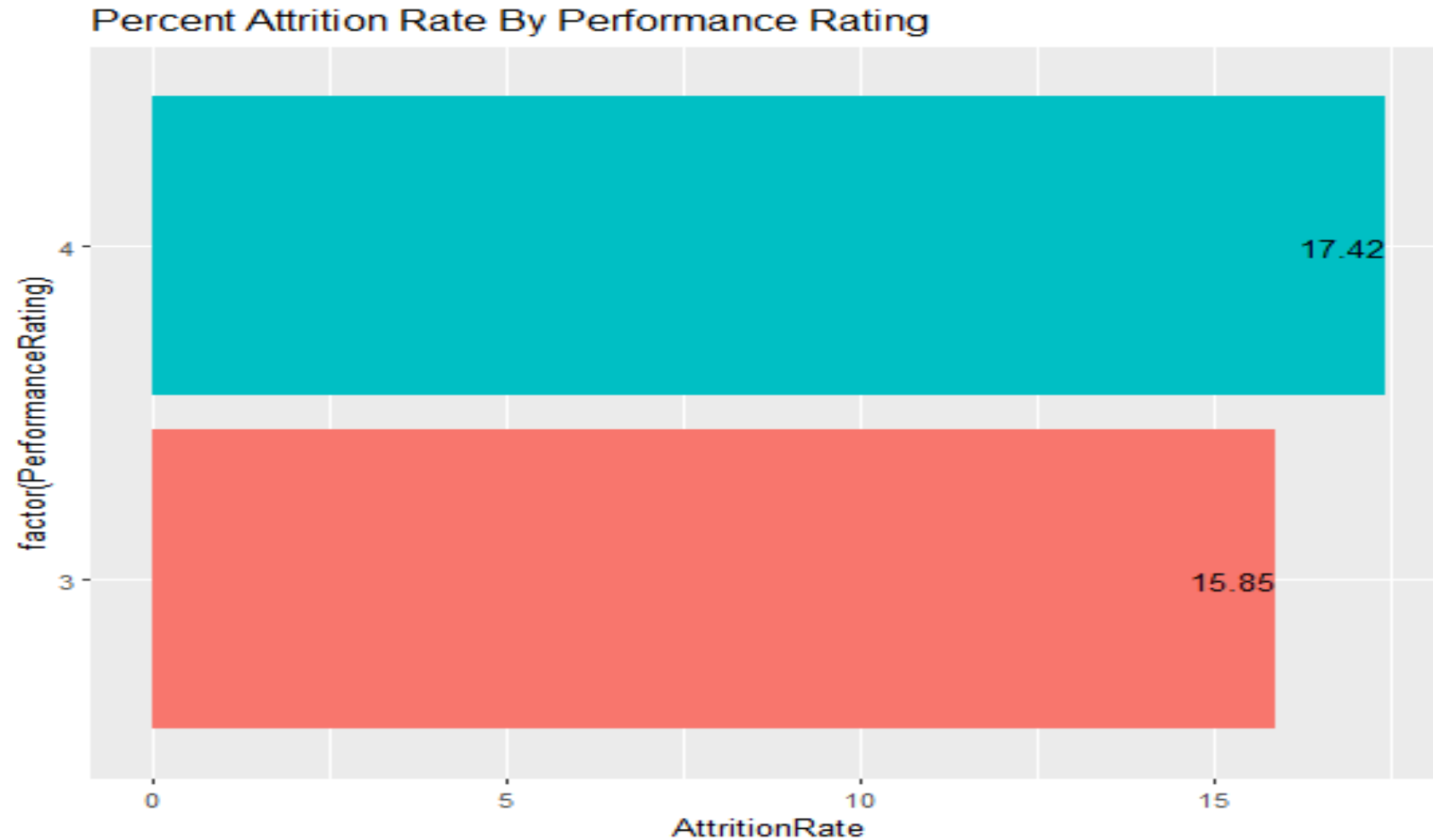
Calculating Attrition Means For Each Factor: Marital Status



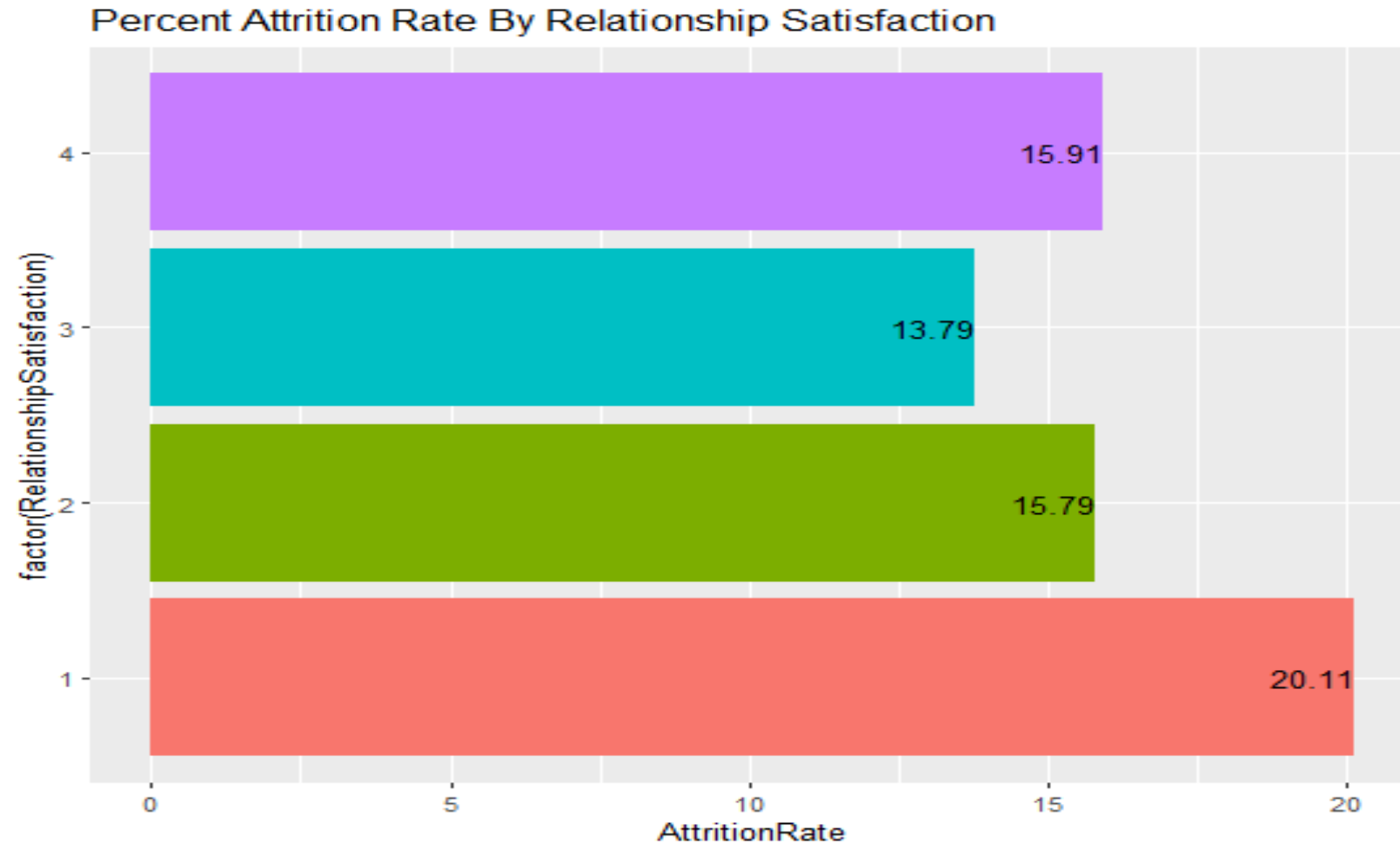
Calculating Attrition Means For Each Factor: Job Satisfaction



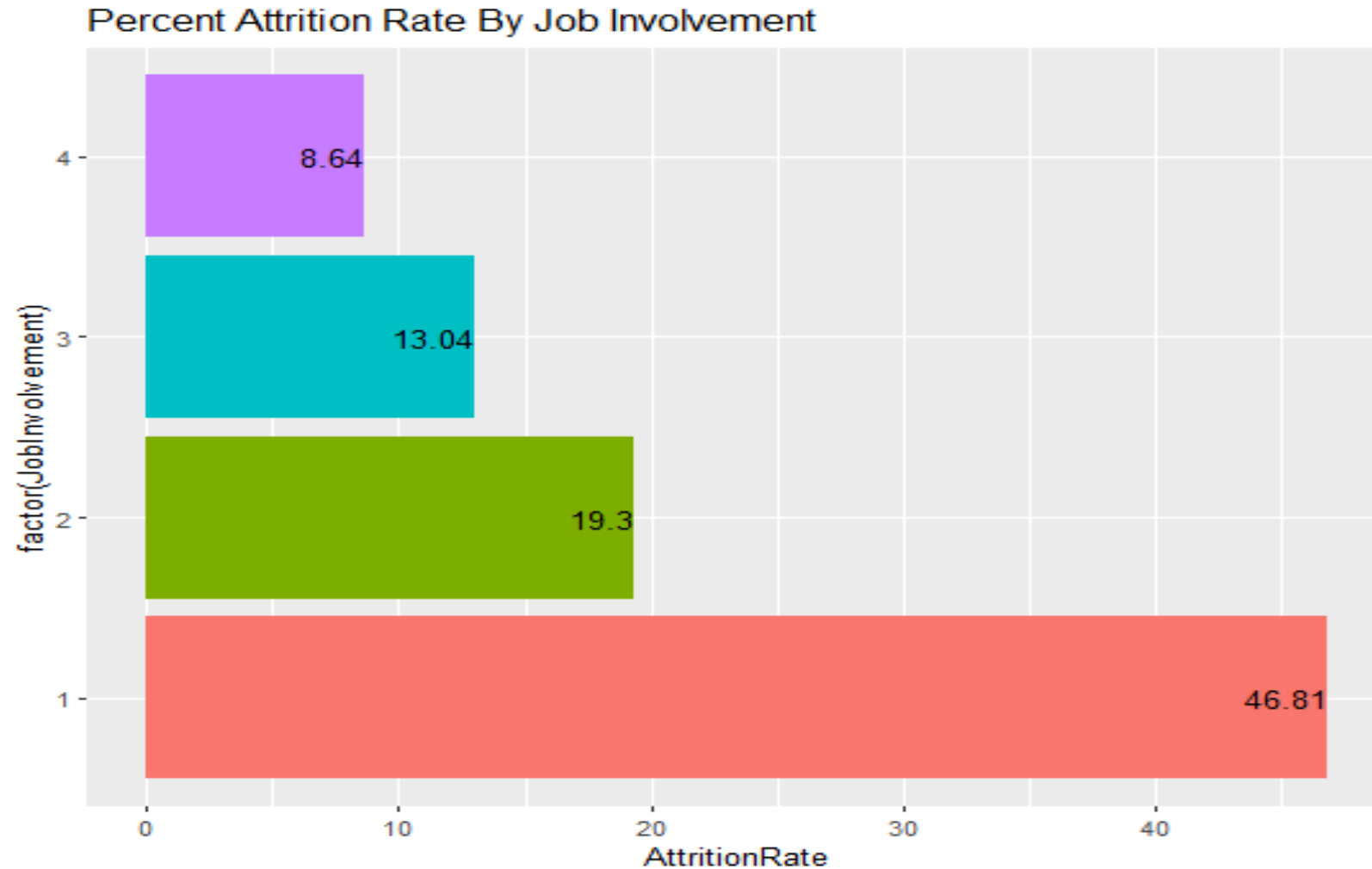
Calculating Attrition Means For Each Factor: Performance Rating



Calculating Attrition Means For Each Factor: Relationship Satisfaction

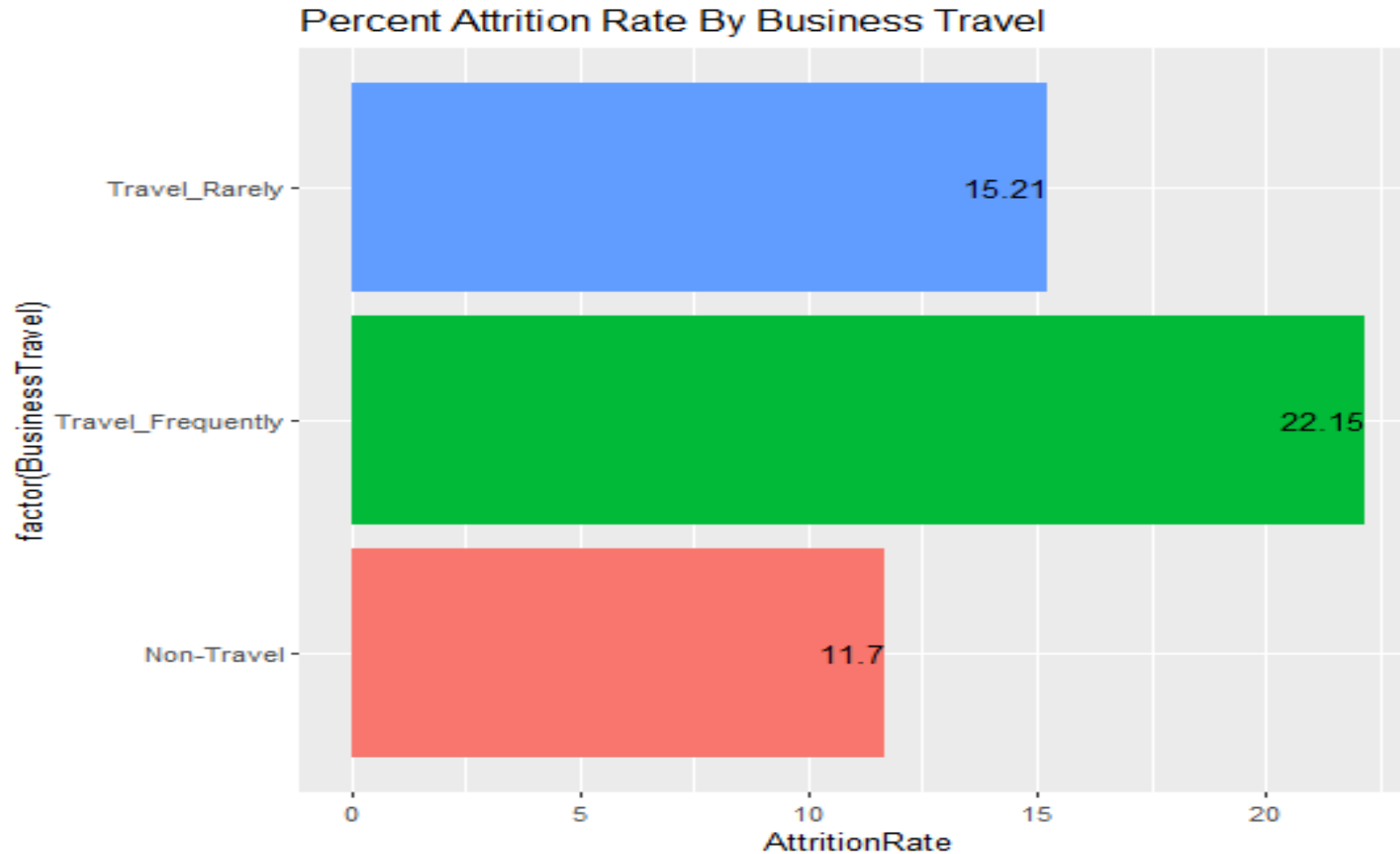


Calculating Attrition Means For Each Factor: Job Involvement

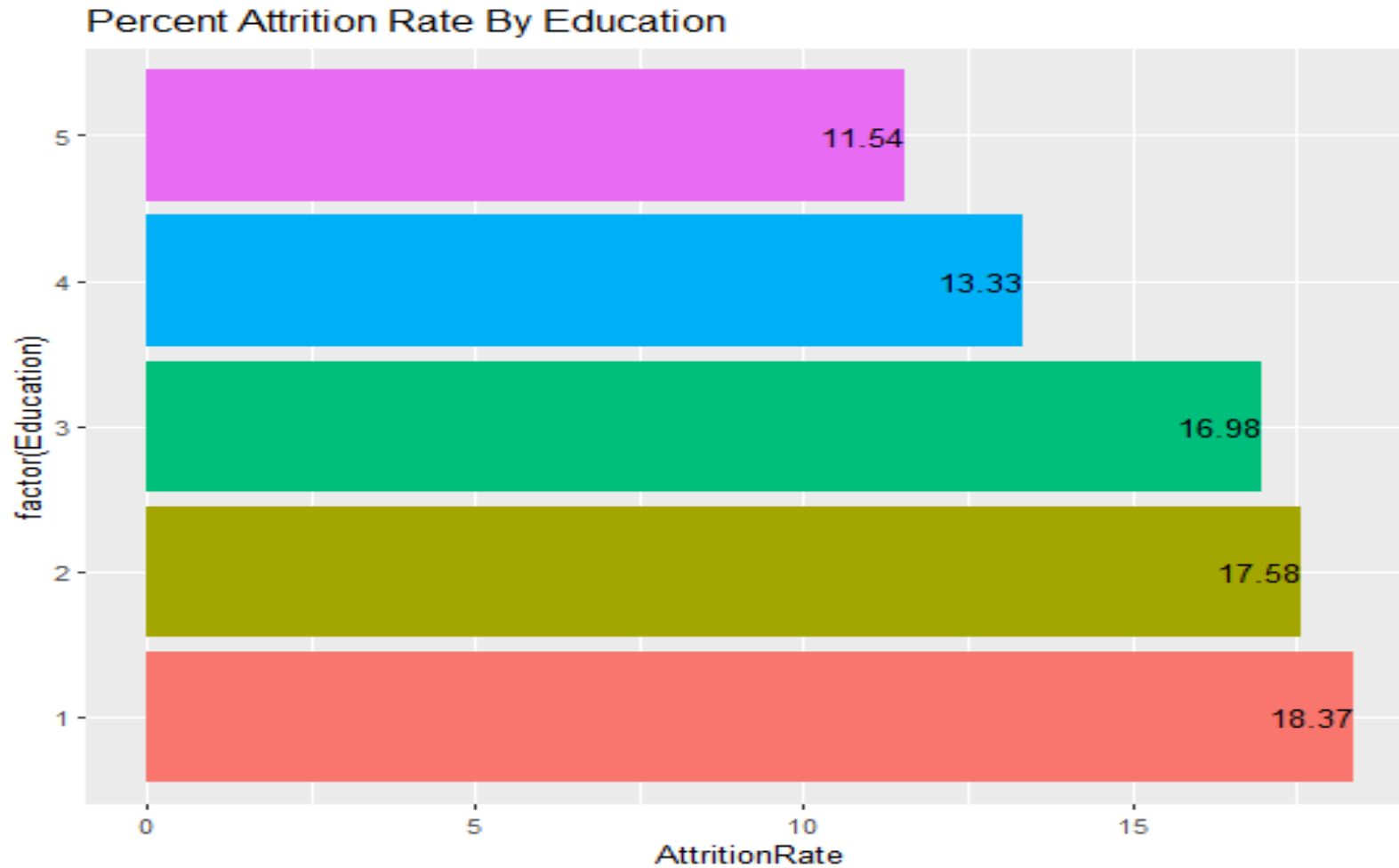


- **Job involvement is a big factor for attrition.**

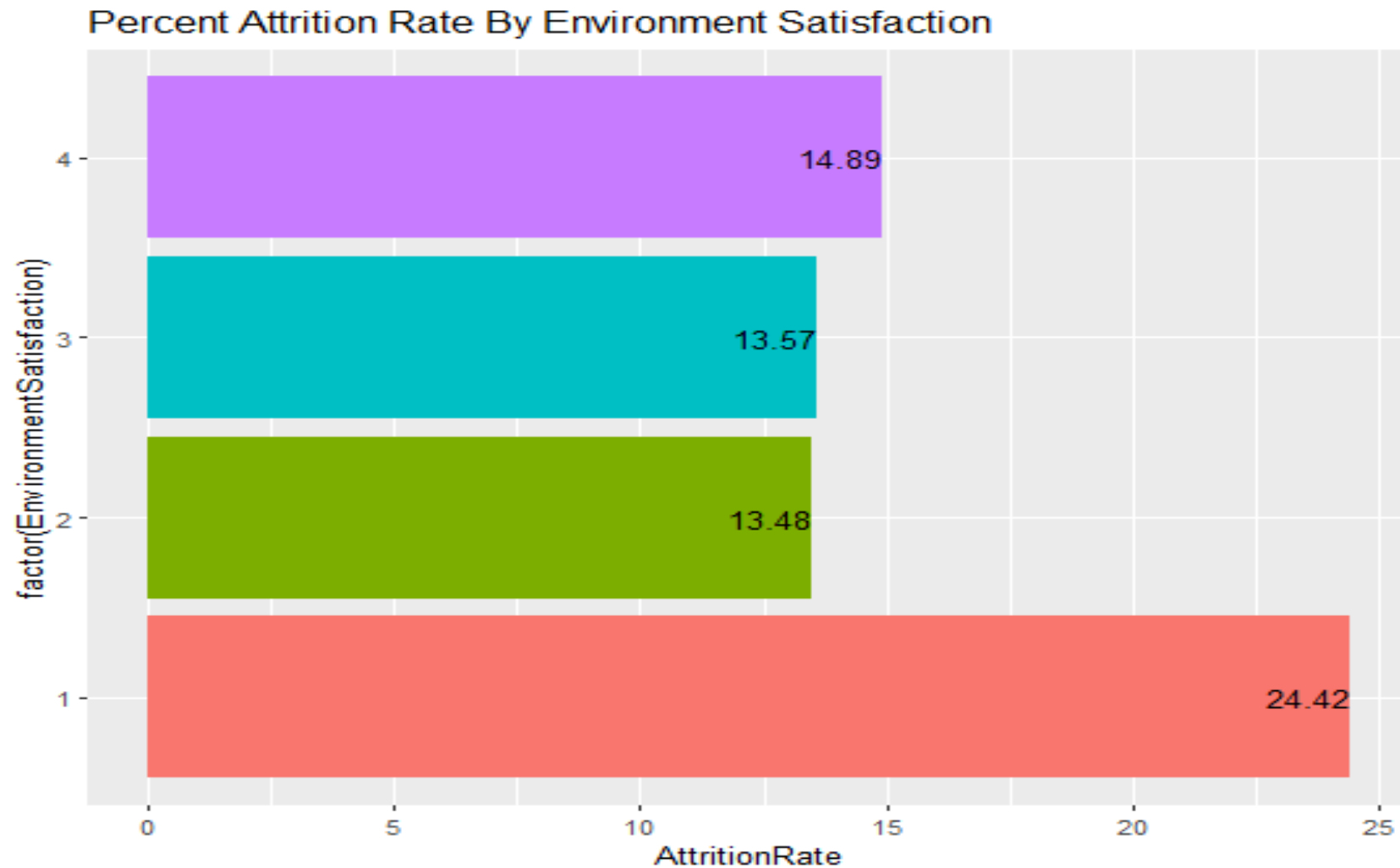
Calculating Attrition Means For Each Factor: Business Travel



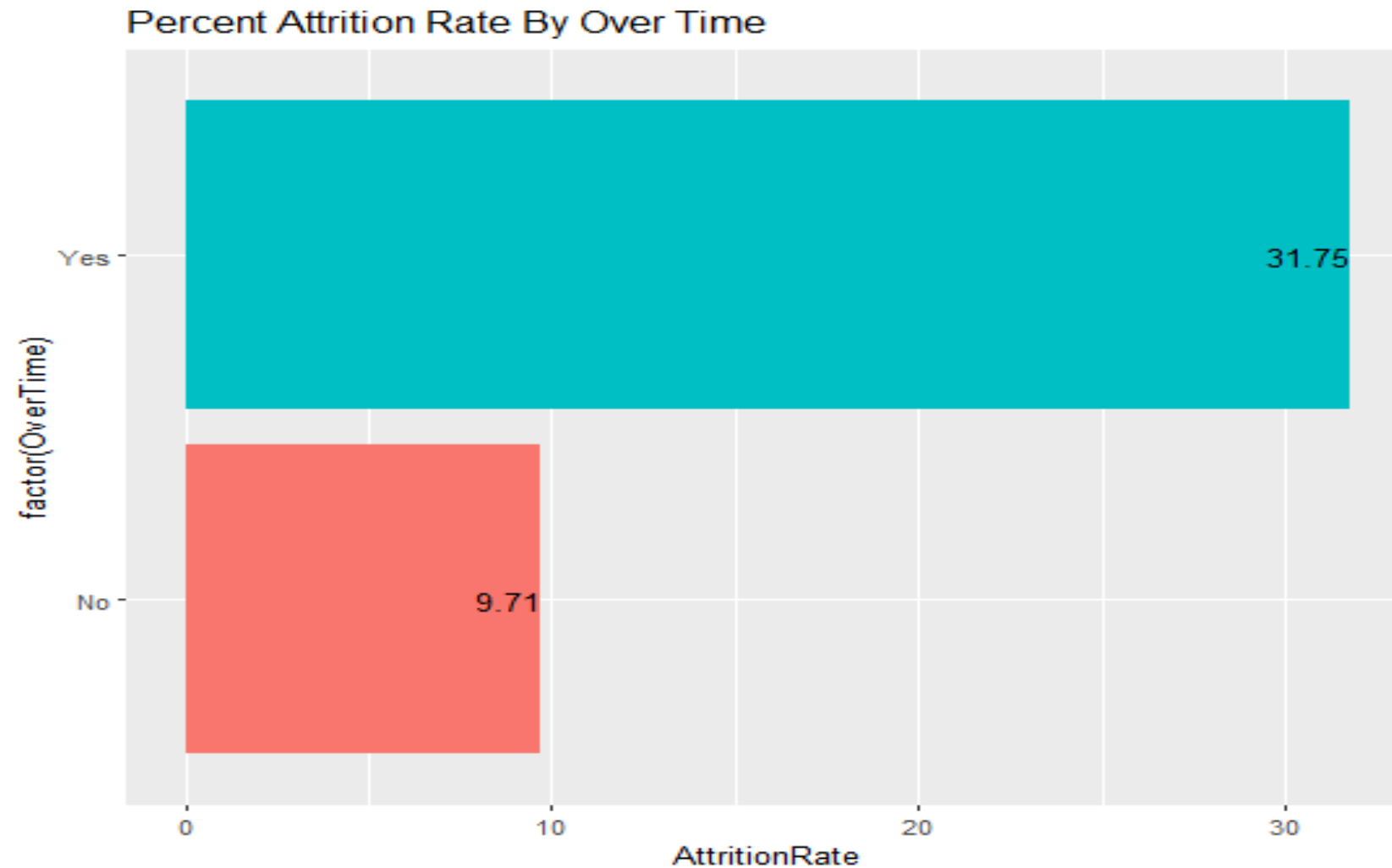
Calculating Attrition Means For Each Factor: Education



Calculating Attrition Means For Each Factor: Environment Satisfaction

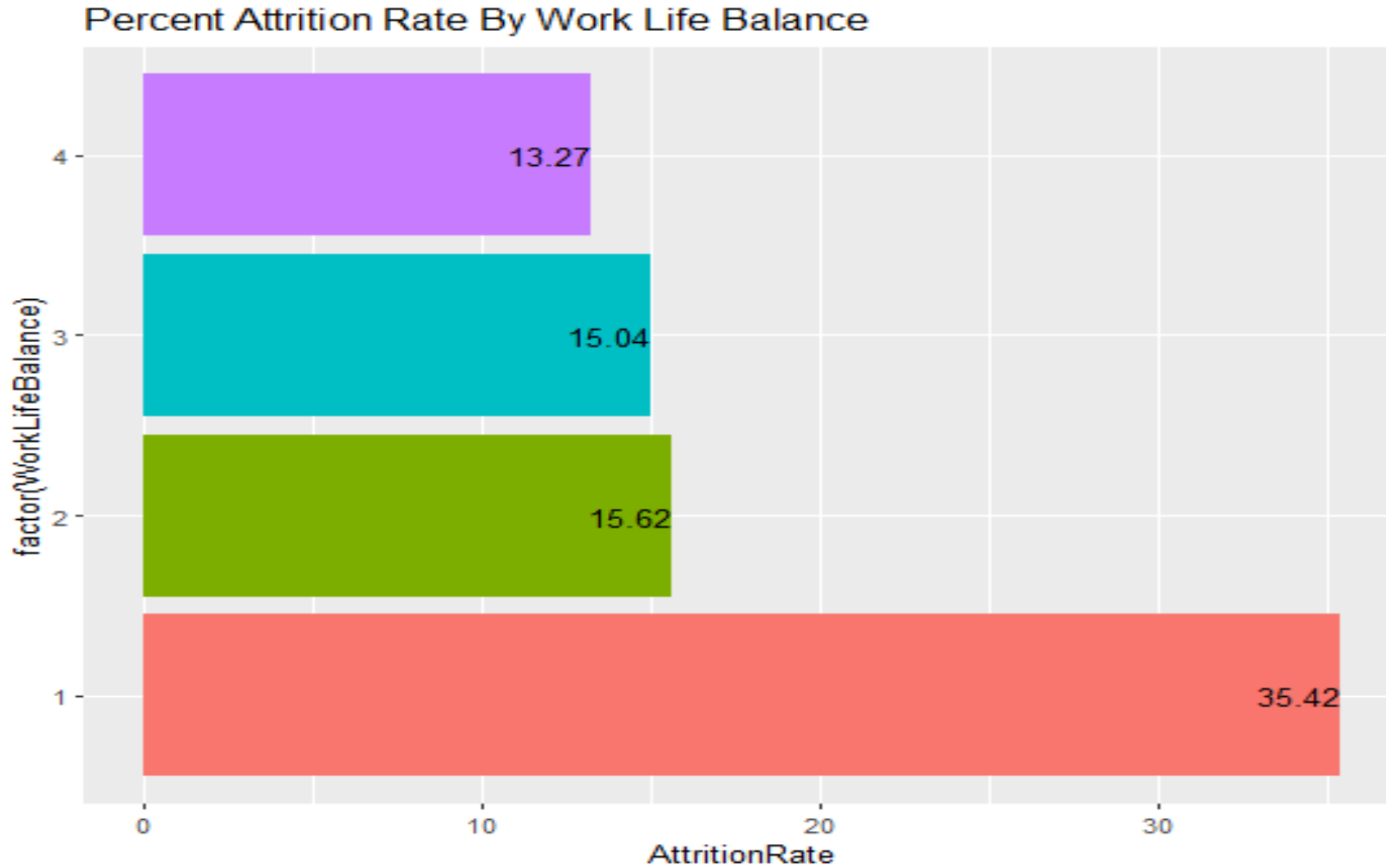


Calculating Attrition Means For Each Factor: Over Time



- **Over Time is a big factor for attrition.**

Calculating Attrition Means For Each Factor: Work Life Balance

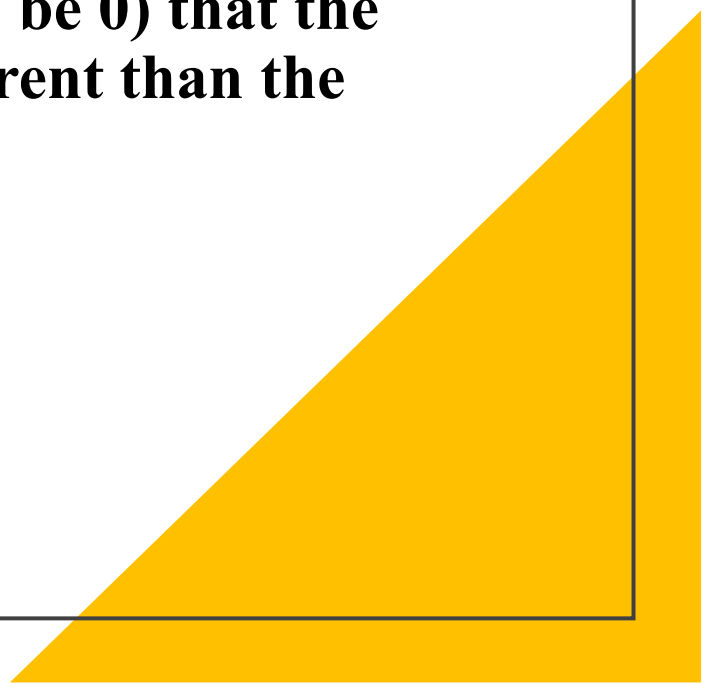


T-Test: Monthly Income and Attrition

- 95% Confidence Interval with $\alpha = 0.05$

Conclusion:

There is sufficient evidence to suggest (p –value estimated to be 0) that the mean monthly income of those that left the company is different than the monthly income of those that stayed at the company.

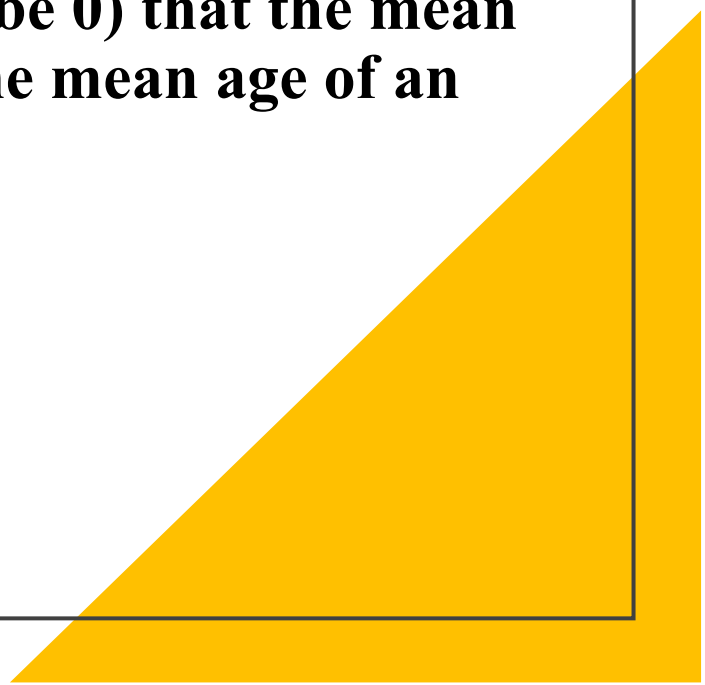
A large yellow right-angled triangle is positioned in the bottom right corner of the slide, pointing towards the top right.

T-Test: Age and Attrition

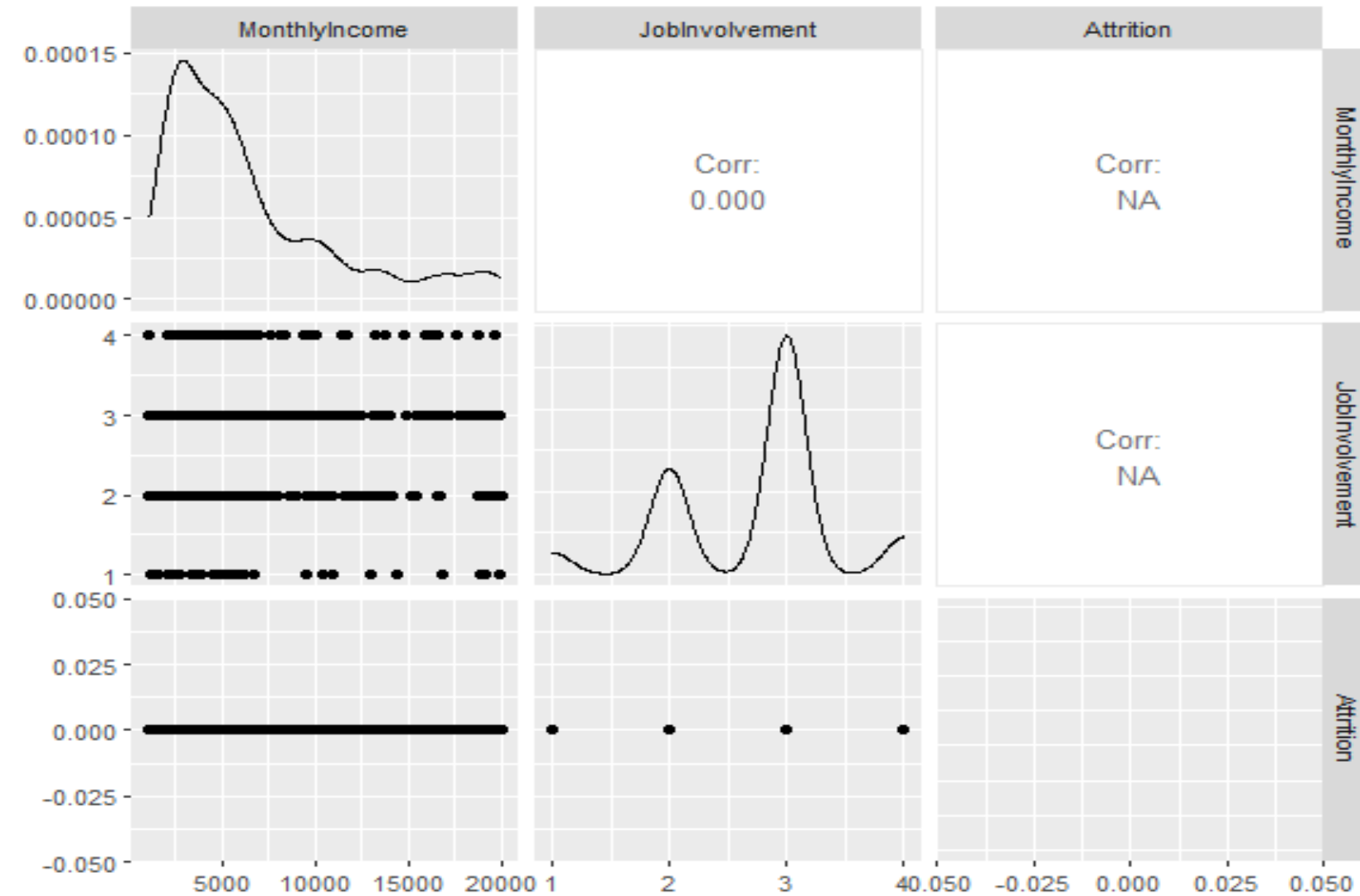
- 95% Confidence Interval with $\alpha = 0.05$

Conclusion:

There is sufficient evidence to suggest (p-value estimated to be 0) that the mean age of an employee who left the company is different than the mean age of an employee who did not leave the company.

A large yellow right-angled triangle is positioned in the bottom right corner of the slide, with its hypotenuse running from the bottom left towards the top right.

Monthly Income and Job Involvement Pair Plot



Naïve Bayes Model 1



Variables:

- Over Time
- Monthly Income
- Attrition

Success (No):

- Employee Stays at The Company

Fail (Yes):

- Employee Leaves The Company

Naïve Bayes Model 1 Prediction Table

Predict Attrition probability for a monthly income of \$5000
for working overtime and not working overtime

Conditional Probabilities

		Monthly Income Predicted	
Actual		No	Yes
	No	\$6,702.00	\$4,675.47
	Yes	\$4,764.79	\$3,786.39

Variables:

- Over Time
- Monthly Income
- Attrition

		Over Time Predicted	
Actual		No	Yes
	No	0.764	0.236
	Yes	0.429	0.571

Success (No):

- Employee Stays at The Company

Fail (Yes):

- Employee Leaves The Company

Naïve Bayes Model 2 (Seed = 7)

Confusion Matrix
Predicted

Actual		No	Yes
	No	216	24
Yes	8	13	

Accuracy:
87.74%

Variables:

- Monthly Income
- Over Time
- Years In Current Role
- Job Involvement

Success (No):

- Employee Stays at The Company

False Positive:
10.00%

Fail (Yes):

- Employee Leaves The Company

False Negative:
38.1%

Naïve Bayes Model 2 100 Test/Train Splits

Confusion Matrix
Predicted

Actual		No	Yes
		No	Yes
No		210	35
Yes		4	12

Accuracy:
85.06%

False Positive:
14.29%

False Negative
25.0%

Variables:

- Monthly Income
- Over Time
- Years In Current Role
- Job Involvement

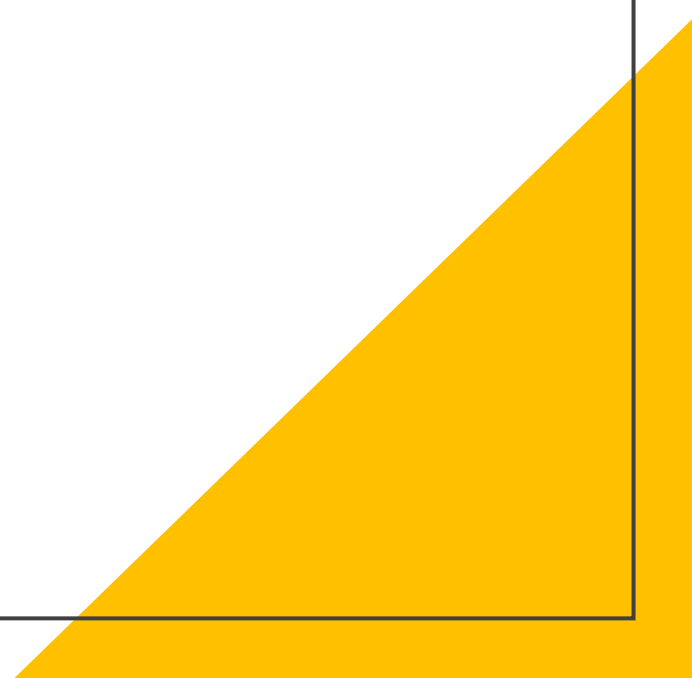
Success (No):

- Employee Stays at The Company

Fail (Yes):

- Employee Leaves The Company

Cost of Replacement and Incentives

- **Entry level replacement cost:**
30% - 50% of employee's annual salary
 - **Mid level employee replacement cost:**
125% - 150% of employee's annual salary
 - **High level employee replacement cost:**
Around 400% of employee's annual salary
 - **Incentives cost:**
\$200 per employee predicted to leave the company
- 
- A large yellow right-angled triangle is positioned in the bottom right corner of the slide, pointing towards the top right.

Naïve Bayes Final Model (Threshold = 0.2 and Seed = 7)

**Confusion Matrix
Predicted**

Actual		No	Yes
	No	143	11
	Yes	81	26

Accuracy:
64.75%

False Positive:
7.14%

False Negative
75.70%

Variables:

- Monthly Income
- Over Time
- Years In Current Role
- Job Involvement
- Years With Current Manager
- Total Working Years
- Age
- Gender

Success (No):

- Employee Stays at The Company

Fail (Yes):

- Employee Leaves The Company

Conclusion

Formula for Calculations Using this Model:

Cost = Sum(Predicted employees to leave the company) * 200

Savings = Sum(Predicted to leave the company and left the company) * Cost of Replacement

Net savings = savings from employees who did not leave - cost of incentive for predicted employees to leave

Using the Naïve Bayes Final Model:

Test Set:

Savings: \$415,220.80

Cost: \$ 20,000.00

Entire Dataset:

Savings: \$ 1,760,542.00

Cost: \$ 69,400.00

Variables:

- Monthly Income
- Over Time
- Years In Current Role
- Job Involvement
- Years With Current Manager
- Total Working Years
- Age
- Gender

Success (No):

- Employee Stays at The Company

Fail (Yes):

- Employee Leaves The Company

KNN Classifier Model 1 (k = 7)

Confusion Matrix

Actual		No	Yes
	No	148	23
	Yes	76	14

Accuracy:
69.35%

False Positive:
13.45%

False Negative:
84.44%

Variables:

- Monthly Income
- Years In Current Role
- Attrition
- Years With Current Manager
- Total Working Years
- Years At Company
- Female

Success (No):

- Employee Stays at The Company

Fail (Yes):

- Employee Leaves The Company

KNN Classifier Model 1 (k = 7)

Confusion Matrix

Predicted

Actual		No	Yes
	No	148	23
	Yes	76	14

Accuracy:
69.35%

Test Set:
Savings: \$ 182,870.40
Cost: \$18,000.00

Success (No):

- Employee Stays at The Company

False Positive:
13.45%

Variables:

- Monthly Income
- Years In Current Role
- Attrition
- Years With Current Manager
- Total Working Years
- Years At Company
- Female

Fail (Yes):

- Employee Leaves The Company

False Negative:
84.44%

KNN Classifier Model 1 ($k = 7$) 100 Train/Test Splits

Confusion Matrix
Predicted

Actual		No	Yes
	No	203	43
	Yes	11	4

Accuracy:
79.31%

False Positive:
14.48%

False Negative:
73.33%

Variables:

- Monthly Income
- Years In Current Role
- Attrition
- Years With Current Manager
- Total Working Years
- Years At Company
- Female

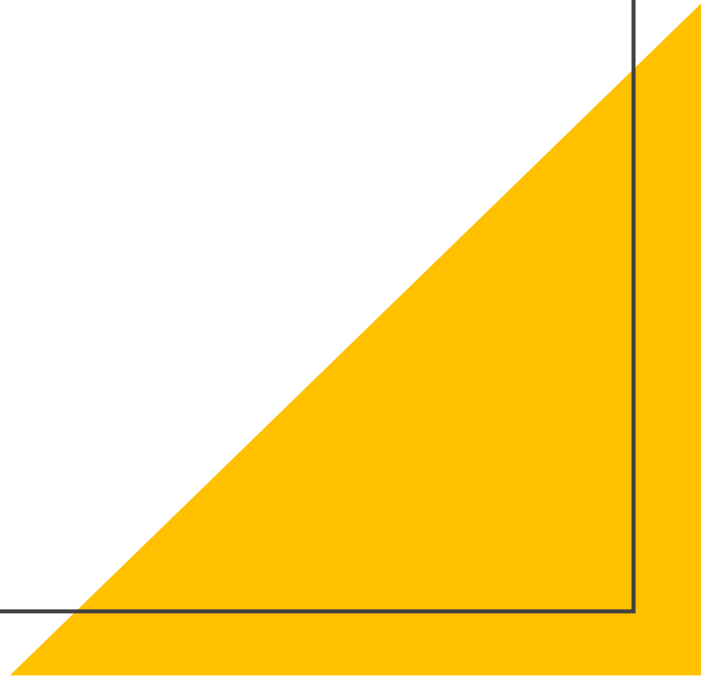
Success (No):

- Employee Stays at The Company

Fail (Yes):

- Employee Leaves The Company

Part 3: Conclusion

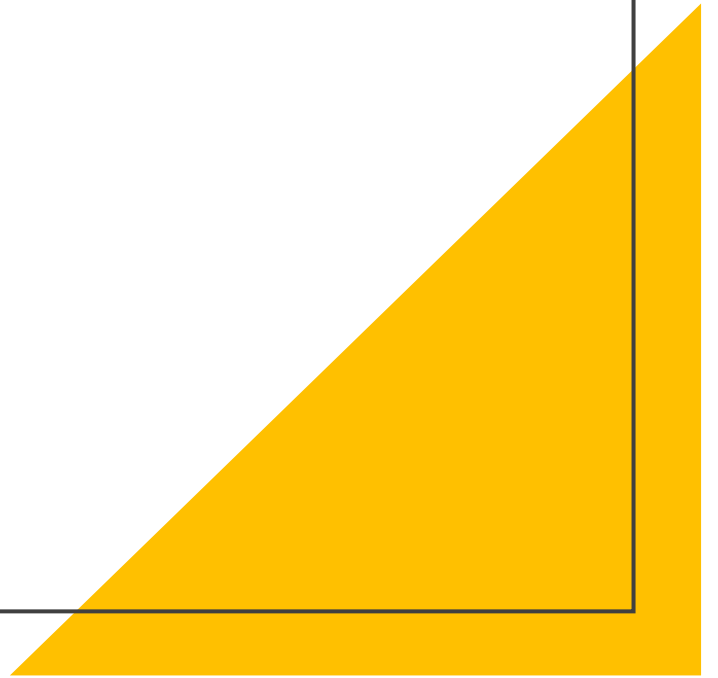


Conclusion

The 3 biggest factors that affect attrition are:

- **Job Involvement**
- **Over Time**
- **Monthly Income**

Another important factor is age.



Conclusion

Formula for Calculations Using this Model:

Cost = Sum(Predicted employees to leave the company) * 200

Savings = Sum(Predicted to leave the company and left the company) * Cost of Replacement

Net savings = savings from employees who did not leave - cost of incentive for predicted employees to leave

Using the Naïve Bayes Final Model:

Test Set:

Savings: \$415,220.80

Cost: \$ 20,000.00

Entire Dataset:

Savings: \$ 1,760,542.00

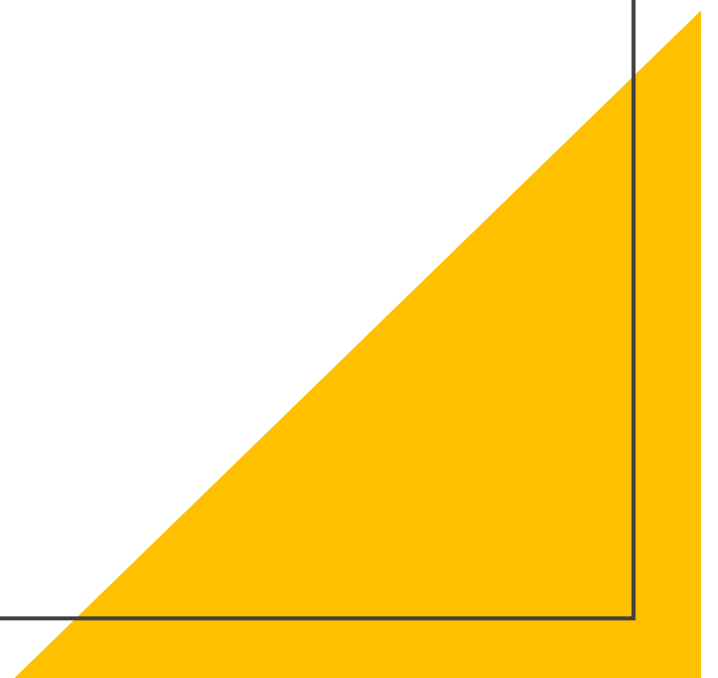
Cost: \$ 69,400.00

Using KNN Classifier k = 7 Model:

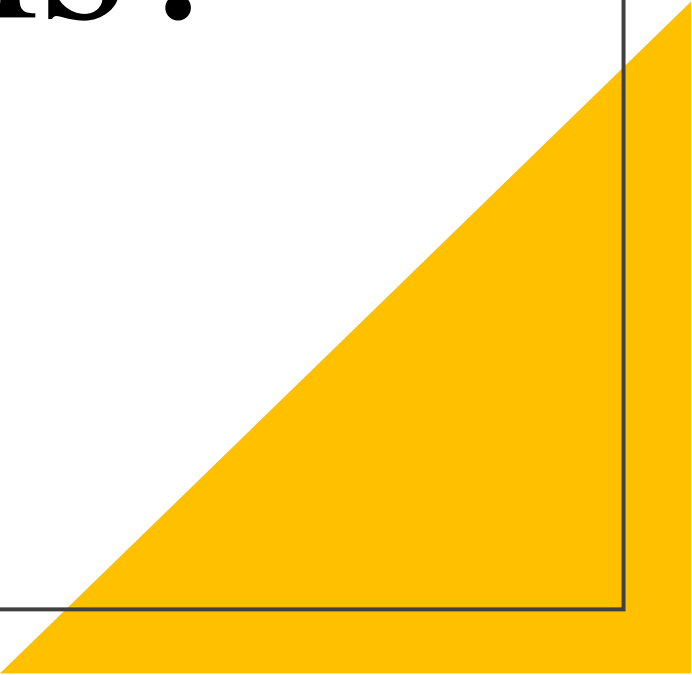
Test Set:

Savings: \$ 182,870.40

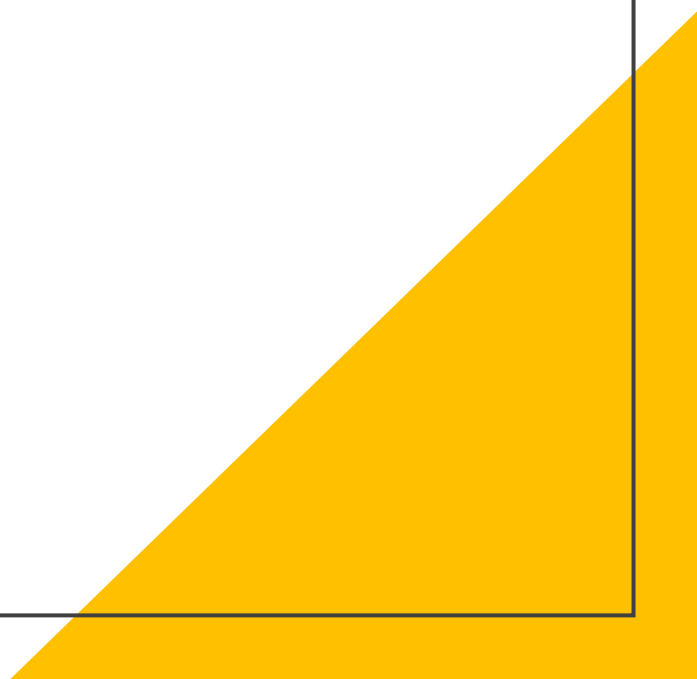
Cost: \$18,000.00



Any Questions?
Thank You

A solid yellow triangle is positioned in the bottom right corner of the slide, pointing towards the top right.

Appendix



T-Test: Monthly Income and Attrition

- 95% confidence level, $\alpha = 0.05$

Step 1: Null Hypothesis and Alternative Hypothesis

- Null Hypothesis: The mean monthly income of an employee who left the company is equal to the mean monthly income an employee who did not leave the company.
- Alternative: Hypothesis: The mean monthly income of an employee who left the company is not equal to the mean monthly income of an employee who did not leave the company.

Step 2 Skip: shading

Step 3: Find the t-test statistic

- The t-statistic is 5.3249 and the degrees of freedom is 228.45
- The 95% confidence interval is (1220.382, 2654.047)

T-Test: Monthly Income and Attrition

Step 4: Find the p-value

- The p-value is $2.412e-07$ which is less than the confidence level (alpha) 0.05.

Step 5: Fail to reject the null hypothesis or reject the null hypothesis

- Reject the null hypothesis.

Step 6: Conclusion

- **Conclusion: There is sufficient evidence to suggest ($p\text{-value} = 2.412e-07$) that the mean monthly income of an employee who left the company is not equal to the mean monthly income of an employee who did not leave the company. This is statistically significant.**